Order of operations with whole numbers

1 Calculate:

a $6 + 4 \times 2$

b $6 + 16 \div 4$

c $24 \div 6 \times 4$

d $22 - 18 \div 6$

e $37 - 35 \div 7 + 8$

f $36 \div 4 \times 8 + 6$

g $18 \div 2 + 7$

h $8 \times 5 - 3 \times 2$

i $7 + 4 \times 3 - 6$

j $11 + 10 \div 5 \times 8$

2 Calculate:

a $(7 + 9) \times 10$

b $7 \times (5 + 4) - 50$

c $34 - (29 + (12 \div 3))$

d $(8 - 2) \times 8$

e $(7 - 2) \times (6 - 2)$

f $(10 + 2) \times (7 - 2)$

g $[69 - (15 + 24)] \div 3 + 8 \times 6$

h $(11 + 5) \div (6 - 2)$

i $[(8 + 2) \times 6] \div 5$

j $\{[100 - (5 + 15)] + 8\} \div 8$

k $[7 \times (6 - 3)] \times [(8 + 2) \div 5]$

l $[1 + (6 \times 3 - 9)] \div [20 - (100 \div 10)]$

3 Consider the expression $4 + 7 \times 3$.

a State whether the following expressions give the same value as the original expression:

i $(4 + 7) \times 3$ **ii** $4 + (7 \times 3)$ **iii** $7 \times 3 + 4$ **iv** $7 + 4 \times 3$

b How do the brackets in $(4 + 7) \times 3$ change the order of operations from the original expression $4 + 7 \times 3$?

4 Consider the expression $7 \div 5 + 9$.

a State whether the following expressions give the same value is the original expression:

i $7 \div (5 + 9)$ **ii** $9 + 7 \div 5$ **iii** $5 + 9 \div 7$ **iv** $(7 \div 5) + 9$

b How do the brackets in $7 \div (5 + 9)$ change the order of operations from the original expression $7 \div 5 + 9$?

5 Consider the expression $5 - 7 \times 8$.

a State whether the following expressions give the same value is the original expression:

i $7 \times 8 - 5$ **ii** $5 - (7 \times 8)$ **iii** $(5 - 7) \times 8$ **iv** $8 \times 5 - 7$

b How do the brackets in $(5 - 7) \times 8$ change the order of operations from the original expression $5 - 7 \times 8$?

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6 Consider the expression $7 \div 4 - 2$.

a State whether the following expressions give the same value is the original expression:

i $2 - 7 \div 4$ **ii** $4 - 2 \div 7$ **iii** $(7 \div 4) - 2$ **iv** $7 \div (4 - 2)$

b How do the brackets in $7 \div (4 - 2)$ change the order of operations from the original expression $7 \div 4 - 2$?

7 Consider the expression $6 + 19 - 4 \times 3$.

a Evaluate $6 + 19 - 4 \times 3$

b The expressions below are like the expression above, except a pair of brackets has been added. Evaluate the following expressions:

i $6 + (19 - 4) \times 3$ **ii** $6 + 19 - (4 \times 3)$

iii $(6 + 19 - 4) \times 3$

8 Consider the expression $6 \times 15 - 5 + 3$.

a Evaluate $6 \times 15 - 5 + 3$

b The expressions below are like the expression above, except a pair of brackets has been added. Evaluate the following expressions:

i $6 \times 15 - (5 + 3)$ **ii** $6 \times (15 - 5) + 3$

iii $(6 \times 15) - 5 + 3$

9 Consider the expression $35 - 6 \times 3 + 2$.

a Evaluate $35 - 6 \times 3 + 2$

b The expressions below are like the expression above, except a pair of brackets has been added. Evaluate the following expressions:

i $35 - (6 \times 3) + 2$ **ii** $35 - (6 \times 3 + 2)$

iii $(35 - 6) \times 3 + 2$ **iv** $35 - 6 \times (3 + 2)$

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